Multi-Task Learning for Stock Price Prediction

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Abstract

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I. Introduction

Insert introduction here

II. Motivation and Objectives

Motivation

Objectives

III. Methodology

A. Data

- Companies
- aggregation choice
- time choice

B. Algorithms

Random Forests: yee

LSTMS: explination

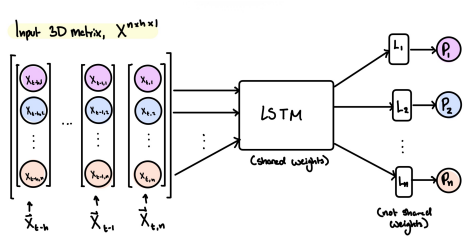

Fig. 1. Diagram of hard parameter sharing and inputs for an LSTM Multi-Tasking model. (Drawn by group).

C. Metrics and Evaluation

D. Preprocessing

Misaligned Dates: Due to different market holidays, some days in the data included both null values and recorded prices. The days with null values were removed to keep the dataset consistent. Dropping a few days of data should not obscure the stocks' growth patterns and relationships.

Logarithmic Scale: The data was scaled logarithmically. This was to ensure stable predictions by reducing effect of outliers on the model and to account for the exponential patterns in each stock's data. The predictions become more stable because most analysis techniques assume additive data, including machine learning. [1].

The scaling affect is shown in appendix A, the medium stock ACIW is on a similar scale to the large stock AAPL after logging the data.

Random Forests: describe script and set up of what it intakes [1]

LSTMS: Pytorch requires LSTM inputs to be 3D tensors. The data looks at stocks, with the only attribute being the stock price. Referring to 1, the matrix on the left would be an example 3D tensor input, with each vector $\vec{x}$ representing all the stocks of interest at the given t. There is no third dimension depicted, as the only attribute of each stock of interest is the given price at time t.

The times within the input are $\vec{x_{t,n}}$ and the lags of $\vec{x_{tn}}$, which represent our rolling window, since $\vec{x_{t+1}}$ should use all the previous lags for its prediction.

E. Building the Models (?)

Random Forests:

LSTMS:

IV. Results

add pics and stuff

A. Hyperparameters

B. Algorithm Performances

V. Discussion

A. Findings

B. Interpretation of Results

C. Hyperparameters

D. Advantages and Disadvantages of Multi-Tasking

VI. Conclusion

References

[1] "Decomposition of time series," 2025. [Online]. Available: https://medium.com/@m.capurso/decomposition-of-time-series-071ce7b7a5a0

Appendix

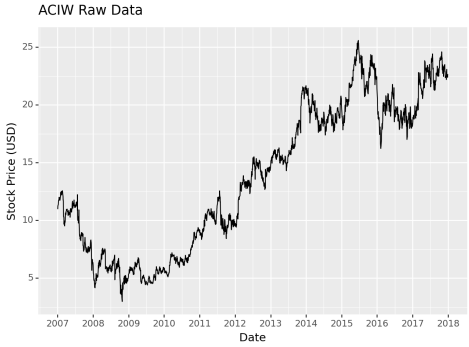

Fig. A2. Raw price data of ACIW Stock

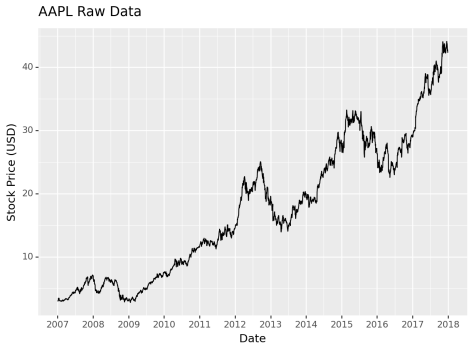

Fig. A3. Raw price data of APPL Stock

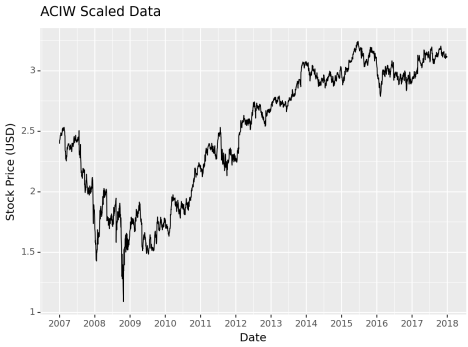

Fig. A4. Logged price data of ACIW Stock

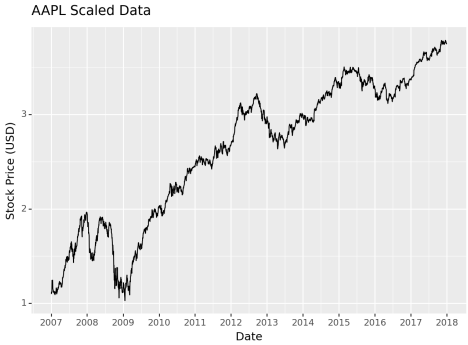

Fig. A5. Logged price data of APPL Stock